

PART A
(15 MARKS)
ANSWER ALL QUESTIONS BELOW

1. Which of the following statement is **TRUE**?

A. $\lim_{x \rightarrow -2} 2 = -2$

B. $\lim_{x \rightarrow -2} x^{-1} = 2$

C. $\lim_{x \rightarrow -2} 2^{-1} = -2$

D. $\lim_{x \rightarrow -2} 2 = 2$

2. Differentiate $\frac{(x+1)^2}{x^2+1}$ with respect to x .

A. $\frac{(x+1)^2(2x) - (x^2+1)(2x+1)}{(x^2+1)^2}$

B. $\frac{(x^2+1)(2x+1) - (x+1)^2(2x)}{(x^2+1)^2}$

C. $\frac{(x^2+1)(2x+2) - (x+1)^2(2x)}{(x^2+1)^2}$

D. $\frac{(x+1)^2(2x) - (x^2+1)(2x+2)}{(x^2+1)^2}$

3. Find $\frac{d^2y}{dx^2}$ for $y = 7x^4 + \frac{8}{x^2}$.

A. $\frac{d^2y}{dx^2} = 84x^2 - 48x^{-4}$

B. $\frac{d^2y}{dx^2} = 84x^2 + 48x^{-4}$

C. $\frac{d^2y}{dx^2} = 84x^3 - 48x^{-3}$

D. $\frac{d^2y}{dx^2} = 84x^3 - 48x^{-2}$

4. If $f(x) = 3x^4 + 4x^3$, which of the following statement is **TRUE**?

A. $(-1, 1)$ is a critical point

B. Critical values are $x = 0$ or $x = -1$

C. Critical points are $(-1, -1)$ and $(0, 0)$

D. $(1, -1)$ is a critical point

5. Determine $\lim_{x \rightarrow 2} \frac{x - 2}{4x^2 - 16}$.

A. $\frac{1}{4}$

B. $\frac{1}{16}$

C. $\frac{1}{8}$

D. $\frac{1}{6}$

6. Which of the following steps below is **TRUE** for the derivative of $(x^2 - x)^3$
- A. $3(x^2 - x)^2(2(x-1))$
- B. $3(x^2 - x)^4(2x-1)$
- C. $3(x^2 - x)^2 2(x-1)$
- D. $3(x^2 - x)^2(2x-1)$
7. Differentiate $\cos^2 3x$ with respect to x .
- A. $6 \cos 3x \sin 3x$
- B. $-2 \cos 3x \sin 3x$
- C. $-6 \cos 3x \sin 3x$
- D. $2 \cos 3x \sin 3x$
8. If the limits for $f(x)$ and $g(x)$ exist at -1 , which of the following is TRUE?
- A. $\lim_{x \rightarrow -1} [f(x) + g(x)] = \lim_{x \rightarrow -1} f(x) + g(x)$
- B. $\lim_{x \rightarrow -1} [f(x) + g(x)] = \lim_{x \rightarrow -1} (f + g)(x)$
- C. $\lim_{x \rightarrow -1} [f(x) + g(x)] = \lim_{x \rightarrow -1} f(x) + \lim_{x \rightarrow -1} g(x)$
- D. $\lim_{x \rightarrow -1} [f(x) + g(x)] = \lim_{x \rightarrow -1} f(x) - \lim_{x \rightarrow -1} g(x)$

9. If $y = \frac{3e^{1-2x}}{4}$, which of the following is the derivative of with respect to x ?

A. $\frac{3e^{-(2x-1)}}{2}$

B. $\frac{-6e^{1-2x}}{4}$

C. $\frac{-3e^{-(2x-1)}}{4}$

D. $\frac{-3e^{2x-1}}{4}$

10. Find y'' for $y = x^4 - 8\sqrt{x}$.

A. $12x^2 + \frac{2}{x\sqrt{x}}$

B. $4x^3 - \frac{4}{\sqrt{x}}$

C. $12x^2 - \frac{4}{\sqrt{x}}$

D. $4x^3 + \frac{4}{\sqrt{x}}$

11. The critical values of $f(x) = 4x^3 - 48x + 24$ are $x = -2$ and $x = 2$. Which of the following critical values correspond to the local minimum?

A. $x = 2$

B. $x = -2$

C. $x = -2$ and $x = 2$.

D. $x = 0$ and $x = 2$.

12. The table below is used to find the local extremum of $y = -x^3 + 3x^2$. Which of the following represents (i), (ii) and (iii).

Interval	$(-\infty, 0)$	$(0, 2)$	$(2, \infty)$
$f'(x)$	$-ve$	(ii)	(iii)
Conclusion	(i)	increasing	decreasing

- A. decreasing, $-ve$, $+ve$
 B. increasing, $-ve$, $+ve$
 C. increasing, $+ve$, $-ve$
 D. decreasing, $+ve$, $-ve$
13. If $\lim_{x \rightarrow 13} f(x) = -3$ and $\lim_{x \rightarrow 13} g(x) = 5$, what is the value of $\lim_{x \rightarrow 13} \frac{(x+7)(f(x)+1)}{g(x)}$.
- A. $\frac{(13+7)(3+1)}{5}$
 B. $\frac{(13+7)(-3+1)}{5}$
 C. $\frac{(13+7)-(3+1)}{5}$
 D. $\frac{(13+7)-(-3+1)}{5}$

14. Find the limit

$$\lim_{x \rightarrow 0} \left(\frac{12}{x} + \frac{3x-12}{x} \right)$$

- A. This limit does not exist.
 B. 1
 C. 12
 D. 3

15. Which of the following derivative is **TRUE**?

A. $\frac{d}{dx}(e^{-3x}) = \frac{3}{-e^{-3x}}$

B. $\frac{d}{dx}(e^{-3x}) = \frac{-3}{e^{-3x}}$

C. $\frac{d}{dx}(e^{-3x}) = \frac{-3}{e^{3x}}$

D. $\frac{d}{dx}(e^{-3x}) = \frac{3}{-e^{-3x}}$

PART B

(25 MARKS)

ANSWER ALL QUESTIONS BELOW

1. Solve for x if $\log_{5-x}(x^2 - 6x + 65) = 2$.

A. 20

B. 10

C. -10

D. -20

2. Simplify $\frac{13}{\sqrt{3}(5 - 2\sqrt{3})}$ in the form $p + q\sqrt{3}$.

A. $3 + \frac{5}{3}\sqrt{3}$

B. $3 - \frac{5}{3}\sqrt{3}$

C. $2 + \frac{5}{3}\sqrt{3}$

D. $2 - \frac{5}{3}\sqrt{3}$

3. Simplify $\frac{(1-a)^{\frac{1}{2}}}{(1-a)^{-\frac{1}{2}} - (1-a)^{\frac{1}{2}}}$.
- A. $\frac{1}{a} + 1$
- B. $\frac{1}{a} - 1$
- C. $1 - \frac{1}{a}$
- D. $-\frac{1}{a} - 1$
4. Differentiate $y = \ln\left(\frac{3-2x}{5x-1}\right)$ with respect to x .
- A. $\frac{2}{3-2x} - \frac{5}{5x-1}$
- B. $\frac{-2}{3-2x} - \frac{5}{5x-1}$
- C. $\frac{-2}{3-2x} + \frac{5}{5x-1}$
- D. $\frac{2}{3-2x} + \frac{5}{5x-1}$
5. Find the value of x such that $(3 - \log_3 x) \log_{3x} 3 = 1$.
- A. $x = -3$
- B. $x = 3$
- C. $x = \frac{1}{3}$
- D. $x = 1$

6. Find the maximum and minimum points for $f(x) = \frac{1}{3}x^3 - x^2 - 8x + 2$.

- A. Maximum point $\left(4, -\frac{74}{3}\right)$ and Minimum point is $\left(-2, \frac{34}{3}\right)$
 B. Minimum point $\left(4, \frac{74}{3}\right)$ and Maximum point is $\left(-2, -\frac{34}{3}\right)$
 C. Minimum point $\left(4, -\frac{74}{3}\right)$ and Maximum point is $\left(-2, \frac{34}{3}\right)$
 D. Maximum point $\left(4, \frac{74}{3}\right)$ and Minimum point is $\left(-2, -\frac{34}{3}\right)$

7. Given that $f(x) = \begin{cases} 3p^4x^2 & , x \leq 1 \\ 3(5p^2 - 4x) & , x > 1 \end{cases}$.

Find the values of p so that $f(x)$ is continuous everywhere.

- A. $p = -1, 1, 2$
 B. $p = 1, -2, 2$
 C. $p = -1, 1, -2, 2$
 D. $p = 1 - 1, -2, 0$
8. The complex number z and w are given by $z = 1 - i$ and $w = \frac{2 - (1 + z)i}{1 + iz}$. Write w as polar form.

- A. $w = \cos\left(\frac{1}{2}\pi\right) + i \sin\left(\frac{1}{2}\pi\right)$
 B. $w = \cos\left(\frac{-3}{2}\pi\right) + i \sin\left(\frac{-3}{2}\pi\right)$
 C. $w = \cos\left(-\frac{\pi}{2}\right) + i \sin\left(-\frac{\pi}{2}\right)$
 D. $w = \cos\left(\frac{3}{2}\pi\right) - i \sin\left(\frac{3}{2}\pi\right)$

9. Find $\frac{dy}{dx}$ for $4x^4 + 10x^3y^2 = y$.

A. $\frac{-16x^3 + 30x^2y^2}{20x^3y + 1}$

B. $\frac{-16x^3 - 30x^2y^2}{20x^3y - 1}$

C. $\frac{-16x^3 - 30x^2y^2}{1 - 20x^3y}$

D. $\frac{16x^3 - 30x^2y^2}{-20x^3y + 1}$

10. A 5.8-foot man is running away from the base of a street light that is 15 feet high.
If he runs at a speed of 23 feet per second, how fast is the length of his shadow changing?

- A. 11.5 feet/second
- B. 12.5 feet/second
- C. 13.5 feet/second
- D. 14.5 feet/second

11. Consider the parametric equations $x = 2t^2$ and $y = 2t^5 - t^2$.

Evaluate $\frac{dy}{dx}$ at point $(2,1)$.

- A. 2
- B. 1
- C. -1
- D. -2

12. If $y = \tan^3(4x)$, evaluate $\frac{dy}{dx}$ when $x = \frac{\pi}{16}$.

- A. 12
- B. 36
- C. 25
- D. 24

13. The function $g(x) = \begin{cases} -2x - 2, & -1 \leq x < \alpha \\ \beta x^2 - 2, & \alpha \leq x < 3 \\ \beta x, & 3 \leq x < 5 \\ -\frac{2}{x} + \beta, & 5 \leq x < 10 \end{cases}$.

Find the values of α and β if $g(x)$ is continuous in the interval $[-1, 3) \cup (3, \infty)$.

- A. $\alpha = 1, \beta = \frac{1}{10}$
- B. $\alpha = 0, \beta = -\frac{1}{10}$
- C. $\alpha = 1, \beta = -\frac{1}{10}$
- D. $\alpha = 0, \beta = \frac{1}{10}$

14. The function g and h are given as $g(x) = \sqrt[3]{x-4}$ and $h(x) = x^3 + 4$. Find the value of x such that $(h \circ g \circ h)(x) = 12$.

- A. 2
- B. 1
- C. -1
- D. -2

15. If $f(x) = \frac{4}{1-x}$, choose the simplified form of $\frac{f(x+h) - f(x)}{h}$.

A. $\frac{-4}{(1-x-h)(1-x)}$

B. $\frac{-4}{(1-x+h)(1-x)}$

C. $\frac{4}{(1-x-h)(1-x)}$

D. $\frac{4}{(1+x+h)(1-x)}$

16. Given that $g(x) = e^{2x} - 3$ and $f \circ g(x) = 2x$. Find $f(x)$.

A. $\ln(x-3)$

B. $\ln(3-x)$

C. $\ln(x+3)$

D. $\frac{1}{2}\ln(u+3)$

17. Solve $\frac{|x+2|}{3|x-4|} \geq 1$

A. $\left(-\frac{5}{2}, 4\right) \cup (4, 7)$

B. $\left[\frac{5}{2}, 4\right) \cup (4, 7]$

C. $\left(\frac{5}{2}, 4\right) \cup (4, 7)$

D. $\left(\frac{5}{2}, 7\right)$

18. The function $f(x) = \begin{cases} \frac{1}{x} & x < \frac{1}{3} \\ 6 - ax & \frac{1}{3} \leq x \end{cases}$ is differentiable at $x = \frac{1}{3}$. What is a ?

A. $a = \frac{1}{-9}$

B. $a = 9$

C. $a = \frac{1}{9}$

D. $a = -9$

19. If $\left| \frac{x+3}{x} \right| > 1$. Which of the following are the steps taken to solve it?

I. $\frac{x+3}{x} - 1 > 0$ or $\frac{x+3}{x} + 1 < 0$

II. $x+3 > x$ or $x+3 < -x$

III. $\frac{x+3}{x} + 1 > 0$ or $\frac{x+3}{x} - 1 < 0$

IV. $\frac{x+3-x}{x} > 0$ or $\frac{x+3+x}{x} < 0$

A. II, III only

B. III, IV only

C. I, IV only

D. I, II only

20. If $x^2 + y^2 = \frac{y}{x}$. Find the expressions of the gradient in terms of x and y at any point (x, y) on the curve.

A. $\frac{dy}{dx} = \frac{3x^2 + y^2}{(2xy - 1)}$

B. $\frac{dy}{dx} = \frac{3x^2 - y^2}{(1 + 2xy)}$

C. $\frac{dy}{dx} = \frac{y^2 - 3x^2}{(1 - 2xy)}$

D. $\frac{dy}{dx} = \frac{3x^2 + y^2}{(1 - 2xy)}$

21. Use the table below. What is the value of $f^2g(-2)$ and $gfg(-1)$?

x	$f(x)$
-3	16
-2	9
-1	4
0	1
1	0
2	1
3	4

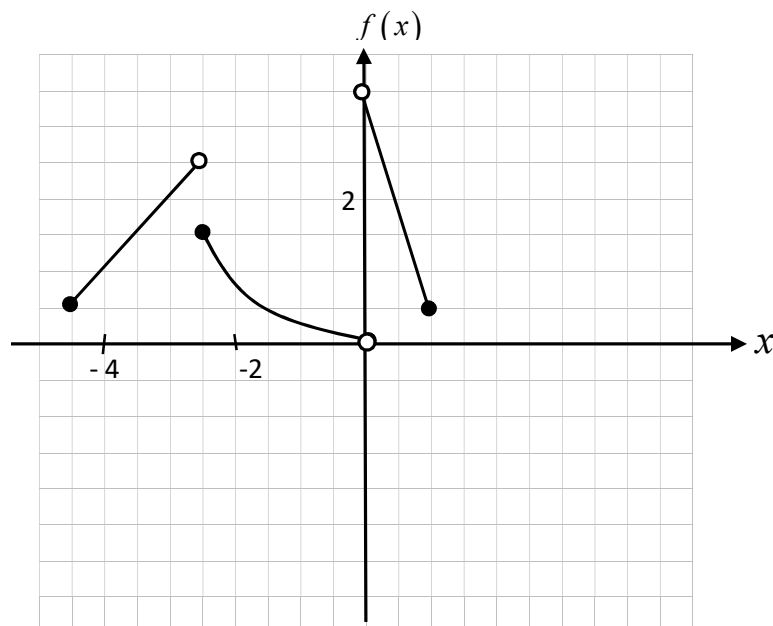
x	$g(x)$
-3	2
-2	1
-1	0
0	-1
1	-2
2	-3
3	-4

- A. 16 and -2
B. 1 and -2
C. 0 and -1
D. 4 and -3

22. If $g(f(x)) = e^x$, and $f(x) = e^{2x}$. Find the function $g(x)$.

- A. $g(x) = x$
- B. $g(x) = \sqrt{2x}$
- C. $g(x) = \sqrt{x}$
- D. $g(x) = \ln x^2$

23. Here is the graph of $y = f(x)$. What are the domain and range of its inverse.



- A. $D_{f^{-1}} = \left(0, \frac{3}{2}\right), R_{f^{-1}} = \left[\frac{-9}{2}, 0\right) \cup (0, 1]$
- B. $D_{f^{-1}} = \left[0, \frac{7}{2}\right), R_{f^{-1}} = \left[\frac{-9}{2}, 0\right) \cup (0, 2]$
- C. $D_{f^{-1}} = \left(0, \frac{7}{2}\right), R_{f^{-1}} = \left[\frac{-9}{2}, 0\right) \cup (0, 1]$
- D. $D_{f^{-1}} = \left[\frac{-9}{2}, 0\right) \cup (0, 1], R_{f^{-1}} = \left(0, \frac{7}{2}\right)$

24. The function g is defined by $g : x \rightarrow (x-1)^2 + 2, x \in [0, \infty)$. Find g^{-1} and state its domain.

- A. $g^{-1}(x) = \sqrt{x+2} + 1, D_{g^{-1}} = [2, \infty)$
B. $g^{-1}(x) = \sqrt{x-2} - 1, D_{g^{-1}} = [-2, \infty)$
C. $g^{-1}(x) = \sqrt{x+2} - 1, D_{g^{-1}} = [-2, \infty)$
D. $g^{-1}(x) = \sqrt{x-2} + 1, D_{g^{-1}} = [2, \infty)$

25. The side of a metal cube, x cm, increases from 9.9 cm at a constant rate of 0.005 cm s^{-1} as it is heated. Find the rate at which the volume is increasing when $t = 20$ seconds.

- A. $1.47 \text{ cm}^3 \text{ s}^{-1}$
B. $1.50 \text{ cm}^3 \text{ s}^{-1}$
C. $0.15 \text{ cm}^3 \text{ s}^{-1}$
D. $1.5 \text{ cm}^3 \text{ s}^{-1}$

END OF QUESTION PAPER